

XTrace & XNysTRace: Optimal Stochastic Trace Estimation

Aaron Guera*

Matthew Martinez†

Tommy Yu‡

Danny Kuei§

1 Introduction

The trace of a matrix $A \in \mathbb{R}^{n \times n}$ is computed via

$$\text{tr}(A) = \sum_{i=1}^n A_{ii}$$

For small matrices, computing this value is trivial; it is simply the sum of diagonals. However, many scientific and machine learning problems in the modern age have to deal with large matrices, simply storing or computing the full matrix may become infeasible due to the computing power or memory issues that it would cause. Examples include Gaussian process models, network analytics, statistical mechanics, and more. These processes are prime examples of where the matrix is too large and only matrix-vector products, also known as matvecs, are practical.

The goal of trace estimation is to estimate $\text{tr}(A)$ in a computationally efficient way to avoid having to deal with these potential problems. Initial papers proposed the Girard-Hutchinson and Hutch++ estimators, with each having their own strengths and weaknesses. The XTrace and XNysTrace estimators are improvements over the Girard-Hutchinson and Hutch++ methods, utilizing variance reduction techniques in tandem with preserving exchangeability, yielding more accurate trace estimates for a fixed computational budget.

1.1 Girard-Hutchinson Estimator

First proposed by Girard and later improved on by Hutchinson, the Girard-Hutchinson estimator uses the isotropic properties of random vectors. Let $\omega \in \mathbb{R}^N$ be a random vector satisfying

$$E[\omega\omega^T] = I$$

Then, for any square matrix $A \in \mathbb{R}^{N \times N}$, through isotropy, it can be shown that

$$E[\omega^T A \omega] = E[\text{tr}(\omega^T A \omega)] = E[\text{tr}(A \omega \omega^T)] = \text{tr}(A E[\omega \omega^T]) = \text{tr}(A)$$

Thus, it is shown that $E[\omega^T A \omega]$ is an unbiased estimator of the trace. Averaging over m independent vectors then yields the Girard-Hutchinson Estimator

$$\hat{\text{tr}}(A) = \frac{1}{m} \sum_{i=1}^m \omega_i^T A \omega_i$$

*Department of Environmental Science and Policy, University of California, Davis, ajguerra@ucdavis.edu

†Department of Statistics, University of California, Davis, mgjmartinez@ucdavis.edu

‡Department of Statistics, University of California, Davis, tyu170@ucdavis.edu

§Department of Statistics, University of California, Davis, dkuei@ucdavis.edu

While this estimator is an unbiased estimator of the trace, the variance converges at a standard Monte-Carlo rate of $\frac{1}{m}$ as we increase the number of samples m .

1.2 Hutch++ Estimator

The Hutch++ estimator improves on the variance convergence rate in comparison to the Girard-Hutchinson estimator. By forming a low rank approximation of the input matrix, the trace of the approximation can be exactly computed as a control variate, making it so that only the trace of the residual needs to be estimated. Let $A \in \mathbb{R}^{N \times N}$ be an input matrix. For a fixed budget m of matvecs, the low rank approximation of A is

$$\hat{A} = QQ^T A$$

Where Q is the orthonormalization of $A[\omega_{m/3+1}, \omega_{m/3+2}, \dots, \omega_{2m/3}]$ and ω_i are iid isotropic vectors and m is divisible by 3. The trace of this low rank approximation is then

$$\text{tr}(\hat{A}) = \text{tr}(QQ^T A) = \text{tr}(Q^T A Q)$$

The Girard-Hutchinson estimator is then applied to the residuals

$$\text{tr}(A - \hat{A}) = \text{tr}((I - QQ^T)A) = \text{tr}((I - QQ^T)A(I - QQ^T))$$

Which produces the result

$$\hat{\text{tr}}_{H++} := \text{tr}(Q^T(AQ)) + \frac{1}{m/3} \sum_{i=1}^{m/3} \omega_i^T (I - QQ^T)(A(I - QQ^T)\omega_i)$$

Even in the worst case scenario, the Hutch++ still has a variance convergence rate of $\frac{1}{m^2}$. While this convergence rate is indeed superior to the Girard-Hutchinson estimator, the Hutch++ loses the principle of exchangeability as a tradeoff.

1.3 Exchangeability

The exchangeability principle states that if the test vectors $\omega_1, \dots, \omega_k$ are exchangeable, then the minimum variance unbiased estimator for $\text{Q}(A)$ is always a symmetric function of $\omega_1, \dots, \omega_k$.

Basically, the family $(\omega_1, \dots, \omega_k)$ has the same distribution as the permuted family of $\omega_{\sigma 1}, \dots, \omega_{\sigma k}$ for every permutation σ in the symmetric group S_k .

In essence, exchangeability means that we do not care about the order in which the random vectors are sampled, guaranteeing that averaging over all permutations preserves unbiasedness and can only reduce variance.

2 Methodology

In this section, we will cover the algorithmic framework of XTrace and XNysTrace estimators, including their efficient implementations, posterior error estimation, and the adaptive stopping mechanism.

2.1 XTrace

2.1.1 Motivation: Limitations of Hutch++

Hutch++ improves the Girard–Hutchinson estimator by forming an even division between the test vectors. The first half of the test vectors is utilized to build a low-rank approximation of the input matrix, while the second half is responsible for estimating the remaining residual trace. Although the variance reduction strategy is effective, it violates the *exchangeability principle*: the estimator is not symmetric over all test vectors, so the role of each vector depends on which side of the split it was designated to. Symmetrizing over all $\binom{m}{m/2}$ would appear to be a sound remedy, however, it would be computationally infeasible due to combinatorial explosion.

2.1.2 Algorithm of XTrace

XTrace resolves the violation of the exchangeability principle committed by Hutch++ via an unbalanced split combined with a Leave-One-Out (LOO) symmetrization. It uses $m/2$ vectors for the low-rank subspace and leaves out exactly one vector to estimate the residual. Let $A \in \mathbb{R}^{N \times N}$ be an input square matrix, m an even number of matrix vector products (matvecs), and $\omega_1, \dots, \omega_{m/2}$ i.i.d. isotropic test vectors forming the test matrix $\Omega = [\omega_1 \cdots \omega_{m/2}]$.

1. **LOO Subspace:** For each $i = 1, \dots, m/2$ drop the i -th column and orthogonalize:

$$Q_{(i)} = \text{orth}(A\Omega_{-i}) :$$

where Ω_{-i} denotes Ω with column i removed. Each ω_i therefore plays a symmetric role.

2. **Basic Estimator:** Each individual estimate decomposes into a low-rank and a residual part:

$$\widehat{\text{tr}}_i = \underbrace{\text{tr}\left(Q_{(i)}^* A Q_{(i)}\right)}_{\text{low-rank part}} + \underbrace{\omega_i^* \left(I - Q_{(i)} Q_{(i)}^*\right) A \left(I - Q_{(i)} Q_{(i)}^*\right) \omega_i}_{\text{residual part}}.$$

3. **XTrace Estimator:** The final estimate is the symmetric average over all LOO estimators:

$$\widehat{\text{tr}}_X = \frac{1}{m/2} \sum_{i=1}^{m/2} \widehat{\text{tr}}_i.$$

By construction, every vector plays a role in both sides of the split, satisfying the exchangeability principle.

2.1.3 Efficient Implementation

A naive implementation would recompute each $Q_{(i)}$ from scratch, incurring a cost of $O(m^2 N)$ per LOO update, which is impractical. XTrace avoids this by implementing the following steps:

1. **Full QR Decomposition:** Compute $A\Omega = QR$ once. This single decomposition drives all subsequent LOO updates.
2. **Rank-One Updates:** Each $Q_{(i)}$ is a rank-one update away from $QQ^\top$, eliminating the need for $O(m^2 N)$ recomputation per LOO step.
3. **Null-Space Solve:** All $s_1, \dots, s_{m/2}$ are obtained simultaneously by solving the singular triangular system $R^\top S = D$.

4. **Final Complexity:** The total cost is $O(m^3)$ for the inversion plus $O(m^2N)$ for the block product—identical to Hutch++. Exchangeability is therefore achieved at a zero additional asymptotic cost.

2.2 XNysTrace

For the special case where $A \in \mathbb{R}^{N \times N}$ is positive semi-definite (PSD), XNysTrace exploits the Nyström approximation to achieve superior performance.

2.2.1 Nyström Approximation

Given a sketch matrix X , the Nyström approximation of A is

$$A\langle X \rangle := AX(X^*AX)^\dagger(AX)^*,$$

which requires only k matvecs. A key structural property is that the Gram matrix formulation allows each LOO update to reduce to a scalar, requiring no recomputation of the subspace.

2.2.2 Key Differences from XTrace

XNysTrace differs from XTrace in two ways. First, it utilizes all m vectors for each LOO approximation, achieving an effective approximation rank of roughly m compared to $m/2$ for XTrace. Second, the higher rank results in a convergence rate approximately $3\times$ faster than Hutch++ for matrices with exponentially decaying eigenvalues. However, it is important to note that there is a primary limitation to XNysTrace. That is, it struggles with slowly decaying eigenvalue spectra, where the nuclear norm penalty in its variance bound becomes significant.

2.3 Posterior Error Estimation and Adaptive Stopping

2.3.1 Posterior Error Estimate

A practical advantage of the LOO structure is that it yields an error estimate free of cost. Reusing the $\ell = m/2$ individual estimators already computed, define:

$$\widehat{\text{err}}^2 = \frac{1}{\ell(\ell-1)} \sum_{i=1}^{\ell} (\widehat{\text{tr}}_i - \widehat{\text{tr}}_X)^2.$$

This estimate requires zero additional matvecs. Its reliability is guaranteed by Proposition 3.1, which shows that the ratio of the error estimate to the true squared error depends only on the correlation between any two individual estimators $\widehat{\text{tr}}_i$.

2.3.2 Adaptive Stopping Criterion

Rather than fixing m in advance, we can set a target relative tolerance $|\text{tr } A - \widehat{\text{tr}}| \leq \varepsilon |\text{tr } A|$. The procedure is as follows:

1. Collect m_0 matvecs $A\omega_1, \dots, A\omega_{m_0}$, setting $j \leftarrow 0$.
2. Form the trace estimate $\widehat{\text{tr}}^{(j)}$ and error estimate $\widehat{\text{err}}^{(j)}$.
3. If $\widehat{\text{err}}^{(j)} \leq \varepsilon |\widehat{\text{tr}}^{(j)}|$: **stop**.
4. Else: collect m_j additional matvecs, set $j \leftarrow j + 1$, $m_j \leftarrow 2m_{j-1}$, and return to step 2.

Each doubling round adds as many products as all prior rounds combined, preserving the $O(m^2N)$ overall complexity. Due to the rapid convergence of XTrace and XNysTrace, only a few doublings are required, capping the total cost at a maximum of $2\times$ the optimal number of matvecs.

3 Theoretical Results

3.1 Variance Bounds

One of the main theoretical guarantees the paper proposes are the variance bounds for the estimators.

Theorem 3.1 (Variance bounds). *Let $\mathbf{A} \in \mathbb{R}^{N \times N}$ be a square matrix and fix $m \geq 8$. The Hutch++, XTrace, and XNysTrace estimators are all unbiased estimators of $\text{tr}(\mathbf{A})$. With standard normal test vectors $\boldsymbol{\omega} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, these estimators satisfy the following variance bounds:*

$$\left(\mathbb{E} |\widehat{\text{tr}}_{H++} - \text{tr } \mathbf{A}|^2\right)^{1/2} \leq \min_{r \leq m/3-2} \sqrt{2} \frac{\|\mathbf{A} - \llbracket \mathbf{A} \rrbracket_r\|_F}{\sqrt{m/3 - r - 1}}, \quad (1)$$

$$\left(\mathbb{E} |\widehat{\text{tr}}_X - \text{tr } \mathbf{A}|^2\right)^{1/2} \leq \sqrt{m} \min_{r \leq m/2-4} \left(\frac{2\|\mathbf{A} - \llbracket \mathbf{A} \rrbracket_r\|}{\sqrt{m/2 - r - 3}} + \frac{2e \|\mathbf{A} - \llbracket \mathbf{A} \rrbracket_r\|_F}{m/2 - r - 3} \right), \quad (2)$$

$$\left(\mathbb{E} |\widehat{\text{tr}}_{XN} - \text{tr } \mathbf{A}|^2\right)^{1/2} \leq m \min_{r \leq m-6} \left(\frac{\sqrt{8}\|\mathbf{A} - \llbracket \mathbf{A} \rrbracket_r\|}{m - r - 5} + \frac{\sqrt{2}\|\mathbf{A} - \llbracket \mathbf{A} \rrbracket_r\|_F}{(m - r - 5)^{3/2}} + \frac{5e^2\|\mathbf{A} - \llbracket \mathbf{A} \rrbracket_r\|_*}{(m - r - 5)^2} \right). \quad (3)$$

Here $\|\cdot\|_F$, $\|\cdot\|$, and $\|\cdot\|_*$ denote the Frobenius, spectral, and nuclear norms respectively, and $\llbracket \mathbf{A} \rrbracket_r$ is the simultaneous best rank- r approximation of $\mathbf{A}$ in these norms.

The bounds involves matrix norms applied to the residual $\mathbf{A} - \llbracket \mathbf{A} \rrbracket_r$ after the best rank- r approximation is removed. When analyzing the difference between the trace estimation methods, the important aspect to look at the norm terms and the differences in the denominator of the Frobenius norm. Hutch++ requires only the Frobenius norm because its split budget design makes the residual estimators conditionally independent given the low-rank approximation. No covariance between estimators arises so only a single Gaussian quadratic form variance calculation is needed. XTrace introduces a second spectral norm term which arises from the covariance between the leave-one-out estimators $\widehat{\text{tr}}_1$ and $\widehat{\text{tr}}_2$. XNysTrace adds a third nuclear norm term which comes from the Nyström approximation error bound. This makes XNysTrace more sensitive to the full spectral tail than randomized SVD bounds which is the price for using only one pass of matvecs as opposed to two for the randomized SVD. Despite having more terms, the Frobenius norm for XTrace and XNysTrace have denominators growing much faster in m , which dominates in practice.

Assumptions One assumption is that the test vectors $\omega_1, \dots, \omega_m$ are drawn i.i.d $\mathcal{N}(\mathbf{0}, \mathbf{I})$. This Gaussian assumption is the most consequential for practical use. The theoretical bounds are derived using properties specific to Gaussian vectors, in particular the identity $\text{Var}[\boldsymbol{\omega}^* \mathbf{M} \boldsymbol{\omega}] = 2\|\mathbf{M}\|_F^2$ for Gaussian $\boldsymbol{\omega}$ and fixed $\mathbf{M}$.

Another assumption is the minimum matvec budget to be $m \geq 8$, which ensures that the denominators in the variance bounds remain positive. In practical applications, this assumption is likely satisfied as these methods are employed for very large values of m .

Finally, we have that the assumption for positive semidefiniteness for XNysTrace. This is because the Nyström approximation itself requires PSD structure to be well-defined and to satisfy the monotonicity property used in the proof.

Consequences As a consequence of Theorem 3.1, we have that XTrace, XNysTrace, and Hutch++ all decrease at a rate of $O(1/m^2)$ which is superior to the Girard-Hutchinson estimator which decreases at the Monte Carlo rate of $O(1/m)$.

We also see how the bound can be an advantage for XTrace and XNysTrace when looking at the eigenvalue decay. Choosing the r proportional to the matvec budget where PSD matrix $\mathbf{A}$ whose eigenvalues have exponential decay with rate $\alpha \in (0, 1)$ gives

$$\begin{aligned} \left(\mathbb{E} |\widehat{\text{tr}}_{H++} - \text{tr } \mathbf{A}|^2 \right)^{1/2} &\leq C_1(\alpha) \alpha^{m/3}, \\ \left(\mathbb{E} |\widehat{\text{tr}}_X - \text{tr } \mathbf{A}|^2 \right)^{1/2} &\leq \sqrt{m} C_2(\alpha) \alpha^{m/2}, \\ \left(\mathbb{E} |\widehat{\text{tr}}_{XN} - \text{tr } \mathbf{A}|^2 \right)^{1/2} &\leq m C_3(\alpha) \alpha^m. \end{aligned}$$

The polynomial prefactors ($\sqrt{m}$ and m) are dominated by the exponential for large m , so XTrace converges at $1.5\times$ and XNysTrace at $3\times$ the rate of Hutch++.

3.2 Proof Summary

In the following, we go through the general steps of the proof on how the variance bounds are obtained. Full proof steps are included in the appendix details.

First, we introduce the following Propositions to establish initial variance bounds for XTrace and XNysTrace estimators.

Proposition 3.1 (XTrace and XNysTrace error). *1. Fix $\mathbf{A} \in \mathbb{R}^{N \times N}$, and consider the XTrace estimator $\widehat{\text{tr}}_X$ with test matrix $\mathbf{\Omega} = [\boldsymbol{\omega}_1 \cdots \boldsymbol{\omega}_{m/2}]$ consisting of $m/2$ standard normal test vectors. The estimator is unbiased: $\mathbb{E} \widehat{\text{tr}}_X = \text{tr } \mathbf{A}$, and the variance satisfies*

$$\text{Var}[\widehat{\text{tr}}_X] \leq \frac{2}{m/2} \mathbb{E} \left\| (\mathbf{I} - \mathbf{Q}_{(1)} \mathbf{Q}_{(1)}^*) \mathbf{A} \right\|_F^2 + 4 \mathbb{E} \left\| (\mathbf{I} - \mathbf{Q}_{(12)} \mathbf{Q}_{(12)}^*) \mathbf{A} \right\|_F^2,$$

where $\mathbf{Q}_{(i)} = \text{orth}(\mathbf{A} \mathbf{\Omega}_{-i})$ and $\mathbf{Q}_{(ij)} = \text{orth}(\mathbf{A} \mathbf{\Omega}_{-ij})$.

2. Let $\mathbf{A} \in \mathbb{R}^{N \times N}$ be psd. Consider the XNysTrace estimator $\widehat{\text{tr}}_{XN}$ with m standard normal test vectors. The estimator is unbiased: $\mathbb{E} \widehat{\text{tr}}_{XN} = \text{tr } \mathbf{A}$, and the variance satisfies

$$\mathbb{E} |\widehat{\text{tr}}_{XN} - \text{tr } \mathbf{A}|^2 \leq \frac{2}{m} \mathbb{E} \left\| \mathbf{A} - \mathbf{A} \langle \mathbf{\Omega}_{-1} \rangle \right\|_F^2 + 2 \mathbb{E} \left\| \mathbf{A} - \mathbf{A} \langle \mathbf{\Omega}_{-12} \rangle \right\|_F^2.$$

To begin, we can use the exchangeability of the test vectors $\boldsymbol{\omega}_1, \dots, \boldsymbol{\omega}_\ell$ ($\ell = m/2$ for XTrace and $\ell = m$ for XNysTrace). For any average of ℓ exchangeable random variables, we have that all individual variances and all pairwise covariances are equal so the variance decomposes as:

$$\text{Var}[\widehat{\text{tr}}] = \underbrace{\frac{1}{\ell} \text{Var}[\widehat{\text{tr}}_1]}_{\text{Term A}} + \underbrace{\left(1 - \frac{1}{\ell}\right) \text{Cov}[\widehat{\text{tr}}_1, \widehat{\text{tr}}_2]}_{\text{Term B}}. \quad (4)$$

3.2.1 Bounding Term A

Denote $\mathbf{\Pi}_i = \mathbf{Q}_{(i)}\mathbf{Q}_{(i)}^*$ and $\mathbf{\Pi}_{ij} = \mathbf{Q}_{(ij)}\mathbf{Q}_{(ij)}^*$. For both methods, the law of total variance and the Gaussian quadratic form identity $\text{Var}[\boldsymbol{\omega}^*\mathbf{M}\boldsymbol{\omega}] = 2\|\mathbf{M}\|_F^2$ give

$$\begin{aligned} \mathbf{A}_X &= \text{Var}[\widehat{\text{tr}}_1] \leq 2\mathbb{E}\|(\mathbf{I} - \mathbf{\Pi}_1)\mathbf{A}(\mathbf{I} - \mathbf{\Pi}_1)\|_F^2, \\ \mathbf{A}_{XN} &= \text{Var}[\widehat{\text{tr}}_1] = 2\mathbb{E}\|\mathbf{A} - \mathbf{A}\langle\boldsymbol{\Omega}_{-1}\rangle\|_F^2. \end{aligned}$$

3.2.2 Bounding Term B

The covariance is harder to bound because the estimators $\widehat{\text{tr}}_1$ and $\widehat{\text{tr}}_2$ are not independent. Both XTrace and XNysTrace use the same algebraic trick to handle this, differing only in the choice of auxiliary matrix. For any random matrix $\mathbf{X}$ independent of both $\boldsymbol{\omega}_1$ and $\boldsymbol{\omega}_2$, we have that

$$\mathbb{E}[(\widehat{\text{tr}}_1 - \text{tr } \mathbf{A} + \text{tr } \mathbf{X} - \boldsymbol{\omega}_1^*\mathbf{X}\boldsymbol{\omega}_1)(\widehat{\text{tr}}_2 - \text{tr } \mathbf{A} + \text{tr } \mathbf{X} - \boldsymbol{\omega}_2^*\mathbf{X}\boldsymbol{\omega}_2)] = \text{Cov}[\widehat{\text{tr}}_1, \widehat{\text{tr}}_2]. \quad (5)$$

This is verified by conditioning sequentially on $\boldsymbol{\Omega}_{-1}$ then $\boldsymbol{\Omega}_{-2}$, exploiting unbiasedness at each step. For both methods, applying Cauchy–Schwarz and exchangeability gives:

$$\mathbf{B} = \text{Cov}[\widehat{\text{tr}}_1, \widehat{\text{tr}}_2] \leq \mathbb{E}|\widehat{\text{tr}}_1 - \text{tr } \mathbf{A} + \text{tr } \mathbf{X} - \boldsymbol{\omega}_1^*\mathbf{X}\boldsymbol{\omega}_1|^2. \quad (6)$$

XTrace Choose $\mathbf{X} = (\mathbf{I} - \mathbf{\Pi}_{12})\mathbf{A}(\mathbf{I} - \mathbf{\Pi}_{12})$, the leave-two-out SVD residual. This matrix is independent of both $\boldsymbol{\omega}_1$ and $\boldsymbol{\omega}_2$ by construction. Expanding (6) using the identity $\text{Var}[\boldsymbol{\omega}^*\mathbf{M}\boldsymbol{\omega}] = 2\|\mathbf{M}\|_F^2$ gives: $\mathbf{B} \leq 2\mathbb{E}\|(\mathbf{I} - \mathbf{\Pi}_1)\mathbf{A}(\mathbf{I} - \mathbf{\Pi}_1) - \mathbf{X}\|_F^2$. Continuing, we solve a chain of norm inequalities, using the fact that orthogonal projectors shrink the Frobenius norm, and that $\mathbf{\Pi}_1 - \mathbf{\Pi}_{12}$ is a rank-one projector yielding $\|(\mathbf{I} - \mathbf{\Pi}_1)\mathbf{A}(\mathbf{I} - \mathbf{\Pi}_1) - \mathbf{X}\|_F^2 \leq 2\|\mathbf{X}\|^2 = 2\|(\mathbf{I} - \mathbf{\Pi}_{12})\mathbf{A}(\mathbf{I} - \mathbf{\Pi}_{12})\|^2$, giving

$$\mathbf{B} \leq 4\mathbb{E}\|(\mathbf{I} - \mathbf{\Pi}_{12})\mathbf{A}(\mathbf{I} - \mathbf{\Pi}_{12})\|^2$$

. Combining A and B gives the variance bound from Proposition 3.1.

XNysTrace Choose $\mathbf{X} = \mathbf{A} - \mathbf{A}\langle\boldsymbol{\Omega}_{-12}\rangle$, the leave-two-out Nyström residual. This is again independent of both $\boldsymbol{\omega}_1$ and $\boldsymbol{\omega}_2$. Expanding (6) gives: $\mathbf{B} \leq 2\mathbb{E}\|\mathbf{A}\langle\boldsymbol{\Omega}_{-12}\rangle - \mathbf{A}\langle\boldsymbol{\Omega}_{-1}\rangle\|_F^2$. The argument now uses a simpler route than XTrace using the fact that $\mathbf{A}\langle\boldsymbol{\Omega}_{-1}\rangle - \mathbf{A}\langle\boldsymbol{\Omega}_{-12}\rangle$ is rank-one and bounded above by $\mathbf{A} - \mathbf{A}\langle\boldsymbol{\Omega}_{-12}\rangle$ in the PSD order. For a PSD rank-one matrix, the Frobenius and spectral norms coincide, and PSD order implies spectral norm order. Therefore $\|\mathbf{A}\langle\boldsymbol{\Omega}_{-1}\rangle - \mathbf{A}\langle\boldsymbol{\Omega}_{-12}\rangle\|_F^2 = \|\mathbf{A}\langle\boldsymbol{\Omega}_{-1}\rangle - \mathbf{A}\langle\boldsymbol{\Omega}_{-12}\rangle\|^2 \leq \|\mathbf{A} - \mathbf{A}\langle\boldsymbol{\Omega}_{-12}\rangle\|^2$, giving

$$\mathbf{B} \leq 2\mathbb{E}\|\mathbf{A} - \mathbf{A}\langle\boldsymbol{\Omega}_{-12}\rangle\|^2$$

. Combining A and B gives the variance bound from Proposition 3.1.

3.2.3 Substituting randomized approximation error bounds

To go from initial variance bounds to complete variance, we use Lemma A.1 which is outlined in the appendix. This step provides deterministic upper bounds on the expected approximation error of the randomized SVD and Nyström approximation.

The XTrace estimator utilizes the randomized SVD bound in which for $r \leq m/2 - 4$,

$$\begin{aligned} \mathbb{E}|\widehat{\text{tr}}_X - \text{tr } \mathbf{A}|^2 &\leq \frac{2}{m/2} \mathbb{E}\left\|(\mathbf{I} - \mathbf{Q}_{(1)}\mathbf{Q}_{(1)}^*)\mathbf{A}\right\|_F^2 + 4\mathbb{E}\left\|(\mathbf{I} - \mathbf{Q}_{(12)}\mathbf{Q}_{(12)}^*)\mathbf{A}\right\|_F^2 \\ &\leq \frac{4m}{m/2 - r - 3} \|\mathbf{A} - [\mathbf{A}]_r\|^2 + \frac{4e^2m}{(m/2 - r - 3)^2} \|\mathbf{A} - [\mathbf{A}]_r\|_F^2. \end{aligned} \quad (7)$$

Taking the square root gives us the final error bound detailed in Theorem 3.1.

XNysTrace utilizes the Nyström approximation and for $r \leq m - 6$,

$$\begin{aligned} \left(\mathbb{E} |\widehat{\text{tr}}_{XN} - \text{tr } \mathbf{A}|^2 \right)^{1/2} &\leq \frac{\sqrt{2}}{\sqrt{m}} \left(\mathbb{E} \|\mathbf{A} - \mathbf{A} \langle \boldsymbol{\Omega}_{-1} \rangle\|_F^2 \right)^{1/2} + \sqrt{2} \left(\mathbb{E} \|\mathbf{A} - \mathbf{A} \langle \boldsymbol{\Omega}_{-12} \rangle\|^2 \right)^{1/2} \\ &\leq \frac{\sqrt{8}m}{m-r-5} \|\mathbf{A} - \llbracket \mathbf{A} \rrbracket_r\| + \frac{\sqrt{2}m}{(m-r-5)^{3/2}} \|\mathbf{A} - \llbracket \mathbf{A} \rrbracket_r\|_F + \frac{5e^2m}{(m-r-5)^2} \|\mathbf{A} - \llbracket \mathbf{A} \rrbracket_r\|_* . \end{aligned} \quad (8)$$

Thus, we confirm the detailed error bounds in Theorem 3.1.

3.2.4 Optimizing over rank r

All that remains is to show that each trace estimator $\widehat{\text{tr}}$ satisfies

$$\left(\mathbb{E} |\widehat{\text{tr}} - \text{tr } \mathbf{A}|^2 \right)^{1/2} \leq \frac{C}{m} \|\mathbf{A}\|_* , \quad (9)$$

for an absolute constant C . We use the following fact: For any matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ and any $r \geq 1$,

$$\|\mathbf{A} - \llbracket \mathbf{A} \rrbracket_r\| \leq \frac{\|\mathbf{A}\|_*}{r+1}, \quad \|\mathbf{A} - \llbracket \mathbf{A} \rrbracket_r\|_F \leq \frac{\|\mathbf{A}\|_*}{2\sqrt{r}} . \quad (10)$$

These follow from the ordering of singular values where σ_{r+1} is at most the average of all singular values, and similarly for the Frobenius norm. Substituting the above fact into the complete bounds and choosing $r = \lfloor m/6 \rfloor - 1$, $r = \lfloor m/4 \rfloor - 2$, and $r = \lfloor m/2 \rfloor - 3$ for Hutch++, X Trace, and XNysTrace respectively balances the numerator decay in r against the denominator growth in $m - r$. This yields the universal bound that all three methods achieve $O(1/m^2)$ variance.

4 Experimental Details

4.1 Setup

In the original manuscript, the authors briefly consider using the XDIAG estimator for network centrality measures that are computed with respect to the diagonal of an adjacency matrix $\mathbf{M}$. While we consider this estimator out of scope for the summary, we examine network centrality measures that are defined in terms of the trace of $\mathbf{M}$. To do so we build four graphs in three families - two Erdős-Rényi (ER) graphs with $c=3$ and $c=10$, a stochastic block model with $k=20$ blocks, and a Barabási-Albert graph with $m=3$ - all with 2000 nodes.

On these graphs we compute and estimate two trace measures: $\text{tr}(M^4)$ and $\text{tr}(e^M)$, which are both PSD. True computation is done by eigendecomposition and $\text{tr}(M^4) = \sum_i \lambda_i^4$ and $\text{tr}(e^M) = \sum_i e^{\lambda_i}$. Estimation is done using the Hutchinson, Hutch++, XTrace, and XNysTrace algorithms, implemented in Python using the efficient $O(m^2N)$ algorithm. Estimation was done at a fixed matvec budget of $m = 100$ with 1000 trials per graph ¹.

For reasons discussed in the Appendix, we expected that the estimation of the two trace measures on the ER graphs will be akin to the flat spectra in the manuscript, which is to say that XTrace will outperform Hutch++ but be outperformed by Hutchinson and XNysTrace. Further we expect that the SBM and BA graphs will be akin to the exponential spectra and errors will be ranked (in descending order) Hutchinson, Hutch++, XTrace, XNysTrace.

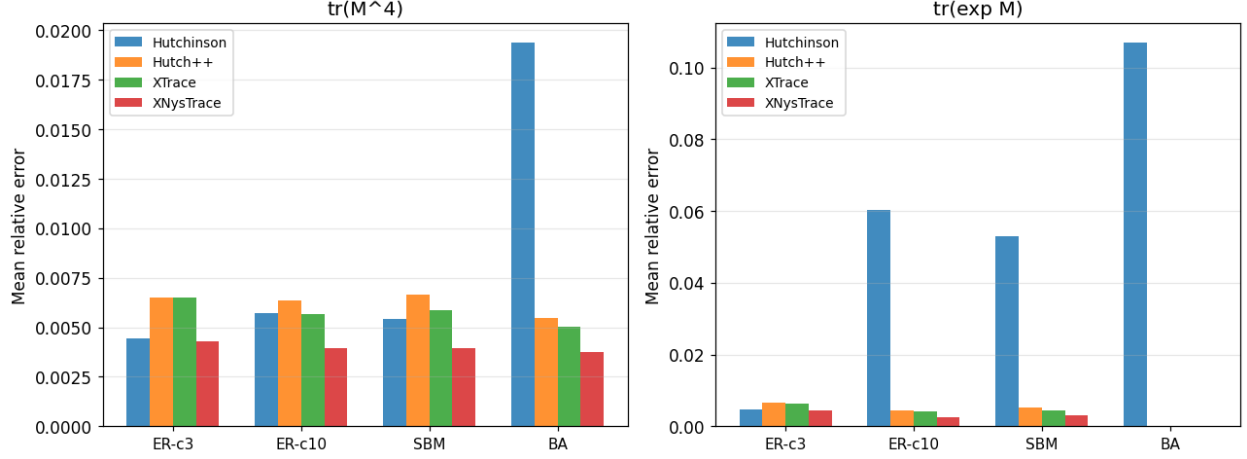


Figure 1: Trace Statistic Error for Simulated Graphs

4.2 Results

In Figure 1 results are plotted for the estimation errors of the four models across two measures and four graphs. In the estimation of $\text{tr}(M^4)$ for ER graphs, we find general support for our hypothesis that XNysTrace and Hutchinson outperform XTrace which in turn outperforms Hutch++. The same can be said for the estimation of $\text{tr}(e^M)$ for ER-c3, but notably as the c parameter grows to 10 we find that the error of the Hutchinson algorithm increases by an order of magnitude. By comparison, the BA graph shows a ranked performance similar to the exponential spectra in the estimation of both $\text{tr}(M^4)$ and in $\text{tr}(e^M)$, only in the latter the three more advanced algorithms have converged².

As expected, the SBM graph shows similarity to the flat spectra in the estimation of $\text{tr}(M^4)$ but to the exponential spectra in the estimation of $\text{tr}(e^M)$, demonstrating that as the eigenvalues are raised from polynomial to exponential powers, the spectral pattern changes and different algorithms may be more appropriate to estimate different network statistics. While these results demonstrate that the XTrace and XNystrace algorithms are generally preferable to the Hutch++ algorithm, in certain cases the Hutchinson algorithm is a competitive alternative.

¹See Figure A1 and the accompanying appendix subsection for further detail and rationale on these choices

²It is likely that setting a lower m parameter for the estimation of $\text{tr}(e^M)$ for the BA graph would show unconverged behavior with the expected performance of Hutchinson-Hutch++-XTrace-XNysTrace

A Additional Proof Details

A.1 Variance Bounds

Unbiased Estimators We can show unbiasedness of XTrace by the fact that each basic estimator $\widehat{\text{tr}}_i$ is conditionally unbiased given $\boldsymbol{\Omega}_{-i}$. The argument proceeds by conditioning on $\boldsymbol{\Omega}_{-i}$, which determines the leave-one-out projector $\boldsymbol{\Pi}_i = \mathbf{Q}_{(i)}\mathbf{Q}_{(i)}^*$. Since $\boldsymbol{\omega}_i$ was not used to construct $\boldsymbol{\Pi}_i$, it is independent of $\boldsymbol{\Pi}_i$ conditional on $\boldsymbol{\Omega}_{-i}$. Applying the isotropy identity $\mathbb{E}[\boldsymbol{\omega}^*\mathbf{M}\boldsymbol{\omega}] = \text{tr}(\mathbf{M})$ for any fixed $\mathbf{M}$:

$$\begin{aligned}\mathbb{E}[\widehat{\text{tr}}_i \mid \boldsymbol{\Omega}_{-i}] &= \mathbb{E}[\text{tr}(\boldsymbol{\Pi}_i\mathbf{A}\boldsymbol{\Pi}_i) + \boldsymbol{\omega}_i^*(\mathbf{I} - \boldsymbol{\Pi}_i)\mathbf{A}(\mathbf{I} - \boldsymbol{\Pi}_i)\boldsymbol{\omega}_i \mid \boldsymbol{\Omega}_{-i}] \\ &= \mathbb{E}[\text{tr}(\boldsymbol{\Pi}_i\mathbf{A}\boldsymbol{\Pi}_i) + \text{tr}((\mathbf{I} - \boldsymbol{\Pi}_i)\mathbf{A}(\mathbf{I} - \boldsymbol{\Pi}_i)) \mid \boldsymbol{\Omega}_{-i}] = \text{tr}(\mathbf{A}).\end{aligned}\quad (11)$$

The final equality uses the fact that the two terms partition $\text{tr}(\mathbf{A})$, which follows by cycling the trace and invoking idempotence of $\mathbf{I} - \boldsymbol{\Pi}_i$. The tower property then gives $\mathbb{E}[\widehat{\text{tr}}_i] = \text{tr}(\mathbf{A})$, and since $\widehat{\text{tr}}_X$ is an average of unbiased estimators, it is itself unbiased.

Bounding Term A The steps to bound Term A for XTrace is as follows

$$\begin{aligned}\mathbf{A} &= \text{Var}[\widehat{\text{tr}}_1] = \mathbb{E}[\text{Var}[\widehat{\text{tr}}_1 \mid \boldsymbol{\Omega}_{-1}]] + \underbrace{\text{Var}[\mathbb{E}[\widehat{\text{tr}}_1 \mid \boldsymbol{\Omega}_{-1}]]}_{=0} \\ &= \mathbb{E}[\text{Var}[\text{tr}(\boldsymbol{\Pi}_1^*\mathbf{A}\boldsymbol{\Pi}_1) + \boldsymbol{\omega}_1^*(\mathbf{I} - \boldsymbol{\Pi}_1)\mathbf{A}(\mathbf{I} - \boldsymbol{\Pi}_1)\boldsymbol{\omega}_1 \mid \boldsymbol{\Omega}_{-1}]] \\ &= \mathbb{E}[\text{Var}[\boldsymbol{\omega}_1^*(\mathbf{I} - \boldsymbol{\Pi}_1)\mathbf{A}(\mathbf{I} - \boldsymbol{\Pi}_1)\boldsymbol{\omega}_1 \mid \boldsymbol{\Omega}_{-1}]] \leq 2\mathbb{E}\|(\mathbf{I} - \boldsymbol{\Pi}_1)\mathbf{A}(\mathbf{I} - \boldsymbol{\Pi}_1)\|_F^2.\end{aligned}$$

Bounding Term B for XTrace The Cauchy-Schwarz bound is given by

$$\begin{aligned}\mathbf{B} &= \text{Cov}[\widehat{\text{tr}}_1, \widehat{\text{tr}}_2] = \mathbb{E}[(\widehat{\text{tr}}_1 - \text{tr} \mathbf{A} + \text{tr} \mathbf{X} - \boldsymbol{\omega}_1^*\mathbf{X}\boldsymbol{\omega}_1)(\widehat{\text{tr}}_2 - \text{tr} \mathbf{A} + \text{tr} \mathbf{X} - \boldsymbol{\omega}_2^*\mathbf{X}\boldsymbol{\omega}_2)] \\ &\leq \mathbb{E}|\widehat{\text{tr}}_1 - \text{tr} \mathbf{A} + \text{tr} \mathbf{X} - \boldsymbol{\omega}_1^*\mathbf{X}\boldsymbol{\omega}_1|^2 \\ &= \mathbb{E}\text{Var}[\boldsymbol{\omega}_1^*[(\mathbf{I} - \boldsymbol{\Pi}_1)\mathbf{A}(\mathbf{I} - \boldsymbol{\Pi}_1) - \mathbf{X}]\boldsymbol{\omega}_1 \mid \boldsymbol{\Omega}_{-1}] \\ &\leq 2\mathbb{E}\|(\mathbf{I} - \boldsymbol{\Pi}_1)\mathbf{A}(\mathbf{I} - \boldsymbol{\Pi}_1) - \mathbf{X}\|_F^2.\end{aligned}$$

Since $\text{range}(\mathbf{Q}_{(12)}) \subseteq \text{range}(\mathbf{Q}_{(1)})$, we have the relations $\boldsymbol{\Pi}_{12} = \boldsymbol{\Pi}_1\boldsymbol{\Pi}_{12}$ and $\mathbf{I} - \boldsymbol{\Pi}_1 = (\mathbf{I} - \boldsymbol{\Pi}_1)(\mathbf{I} - \boldsymbol{\Pi}_{12})$. Since $\mathbf{X} = (\mathbf{I} - \boldsymbol{\Pi}_{12})\mathbf{A}(\mathbf{I} - \boldsymbol{\Pi}_{12})$, it follows that

$$\begin{aligned}\|(\mathbf{I} - \boldsymbol{\Pi}_1)\mathbf{A}(\mathbf{I} - \boldsymbol{\Pi}_1) - \mathbf{X}\|_F^2 &= \|(\mathbf{I} - \boldsymbol{\Pi}_1)\mathbf{X}(\mathbf{I} - \boldsymbol{\Pi}_1) - \mathbf{X}\|_F^2 \\ &= \|(\mathbf{I} - \boldsymbol{\Pi}_1)\mathbf{X}\boldsymbol{\Pi}_1 + \boldsymbol{\Pi}_1\mathbf{X}\|_F^2 \\ &= \|(\mathbf{I} - \boldsymbol{\Pi}_1)\mathbf{X}\boldsymbol{\Pi}_1\|_F^2 + \|\boldsymbol{\Pi}_1\mathbf{X}\|_F^2 \\ &\leq \|\mathbf{X}\boldsymbol{\Pi}_1\|_F^2 + \|\boldsymbol{\Pi}_1\mathbf{X}\|_F^2 \\ &= \|\mathbf{X}(\boldsymbol{\Pi}_1 - \boldsymbol{\Pi}_{12})\|_F^2 + \|(\boldsymbol{\Pi}_1 - \boldsymbol{\Pi}_{12})\mathbf{X}\|_F^2 \\ &= \|\mathbf{X}(\boldsymbol{\Pi}_1 - \boldsymbol{\Pi}_{12})\|^2 + \|(\boldsymbol{\Pi}_1 - \boldsymbol{\Pi}_{12})\mathbf{X}\|^2 \leq 2\|\mathbf{X}\|^2.\end{aligned}$$

The Pythagorean theorem is used to pass to the third line while the fifth line uses the fact that $\mathbf{X} = \mathbf{X}(\mathbf{I} - \boldsymbol{\Pi}_{12})$ and $\mathbf{X} = (\mathbf{I} - \boldsymbol{\Pi}_{12})\mathbf{X}$. For the sixth line, note $\boldsymbol{\Pi}_1 - \boldsymbol{\Pi}_{12}$ is a rank-one orthogonal projector, the Frobenius norm and spectral norm coincide for rank-one matrices. Combining the last two displays, we deduce that

$$\mathbf{B} = \text{Cov}[\widehat{\text{tr}}_1, \widehat{\text{tr}}_2] \leq 4\mathbb{E}\|(\mathbf{I} - \boldsymbol{\Pi}_{12})\mathbf{A}(\mathbf{I} - \boldsymbol{\Pi}_{12})\|^2.$$

Bounding Term B for XNysTrace The Cauchy-Schwarz bound is given by

$$\begin{aligned}
B &= \text{Cov} [\widehat{\text{tr}}_1, \widehat{\text{tr}}_2] = \mathbb{E} [(\widehat{\text{tr}}_1 - \text{tr } \mathbf{A} + \text{tr } \mathbf{X} - \boldsymbol{\omega}_1^* \mathbf{X} \boldsymbol{\omega}_1)(\widehat{\text{tr}}_2 - \text{tr } \mathbf{A} + \text{tr } \mathbf{X} - \boldsymbol{\omega}_2^* \mathbf{X} \boldsymbol{\omega}_2)] \\
&\leq \mathbb{E} |\widehat{\text{tr}}_1 - \text{tr } \mathbf{A} + \text{tr } \mathbf{X} - \boldsymbol{\omega}_1^* \mathbf{X} \boldsymbol{\omega}_1|^2 \\
&= \mathbb{E} \text{Var} [\boldsymbol{\omega}_1^* [\mathbf{A} \langle \boldsymbol{\Omega}_{-12} \rangle - \mathbf{A} \langle \boldsymbol{\Omega}_{-1} \rangle] \boldsymbol{\omega}_1 \mid \boldsymbol{\Omega}_{-1}] \\
&= 2 \mathbb{E} \|\mathbf{A} \langle \boldsymbol{\Omega}_{-1} \rangle - \mathbf{A} \langle \boldsymbol{\Omega}_{-12} \rangle\|_F^2.
\end{aligned}$$

The psd matrix $\mathbf{A} \langle \boldsymbol{\Omega}_{-1} \rangle - \mathbf{A} \langle \boldsymbol{\Omega}_{-12} \rangle$ has rank one, and it is bounded above by $\mathbf{A} - \mathbf{A} \langle \boldsymbol{\Omega}_{-12} \rangle$ in the psd order. Therefore,

$$B = \text{Cov} [\widehat{\text{tr}}_1, \widehat{\text{tr}}_2] \leq 2 \mathbb{E} \|\mathbf{A} - \mathbf{A} \langle \boldsymbol{\Omega}_{-12} \rangle\|^2.$$

Substituting randomized approximation error bounds

Lemma A.1 (Randomized SVD and randomized Nyström error). *Fix a matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$, and draw a standard normal matrix $\boldsymbol{\Omega} \in \mathbb{R}^{N \times k}$. For any $r \leq k - 2$, the randomized SVD error is bounded by*

$$\mathbb{E} \|(\mathbf{I} - \mathbf{Q}\mathbf{Q}^*)\mathbf{A}\|^2 \leq \frac{k+r-1}{k-r-1} \left(\|\mathbf{A} - \llbracket \mathbf{A} \rrbracket_r\|^2 + \frac{e^2}{k-r} \|\mathbf{A} - \llbracket \mathbf{A} \rrbracket_r\|_F^2 \right), \quad (12)$$

$$\mathbb{E} \|(\mathbf{I} - \mathbf{Q}\mathbf{Q}^*)\mathbf{A}\|_F^2 \leq \frac{k-1}{k-r-1} \|\mathbf{A} - \llbracket \mathbf{A} \rrbracket_r\|_F^2, \quad (13)$$

where $\mathbf{Q} = \text{orth}(\mathbf{A}\boldsymbol{\Omega})$.

Assume that $\mathbf{A} \in \mathbb{R}^{N \times N}$ is a psd matrix. For any $r \leq k - 4$, the randomized Nyström error is bounded by

$$(\mathbb{E} \|\mathbf{A} - \mathbf{A} \langle \boldsymbol{\Omega} \rangle\|^2)^{1/2} \leq \frac{k+r-1}{k-r-3} \left(\|\mathbf{A} - \llbracket \mathbf{A} \rrbracket_r\| + \frac{\sqrt{3}e^2}{k-r} \|\mathbf{A} - \llbracket \mathbf{A} \rrbracket_r\|_* \right), \quad (14)$$

$$(\mathbb{E} \|\mathbf{A} - \mathbf{A} \langle \boldsymbol{\Omega} \rangle\|_F^2)^{1/2} \leq \frac{k-2}{k-r-3} \left(\|\mathbf{A} - \llbracket \mathbf{A} \rrbracket_r\|_F + \frac{1}{\sqrt{k-r}} \|\mathbf{A} - \llbracket \mathbf{A} \rrbracket_r\|_* \right). \quad (15)$$

B Additional Experimental Details

B.0.1 Reproduction of Spectral Simulation

Here, in Figure A1 we present our reproduction of Figure 3 from the original manuscript with the intention of rationalizing our choices of graphs and parameters in the experimental extension. Our reproduction differs from the original only in that we estimate 100 trials per point instead of 1000. This was due to computational feasibility - even at 100 trials, computation time was near 1000 seconds for all spectral profiles. As such we may see more MC noise in each figure but no other structural or significant differences.

With respect to our selection for m , observe in the exponential spectra at $m = 100$ that the estimators are well differentiated in terms of error. Thus in our network extensions we parameterize accordingly to identify exponential-like spectra and their performance with trace-based statistics.

With respect to the selection of the graph models, we identified graphs with spectra similar to the flat and exponential spectra. Erdős-Rényi graphs have equal expected degree for all nodes

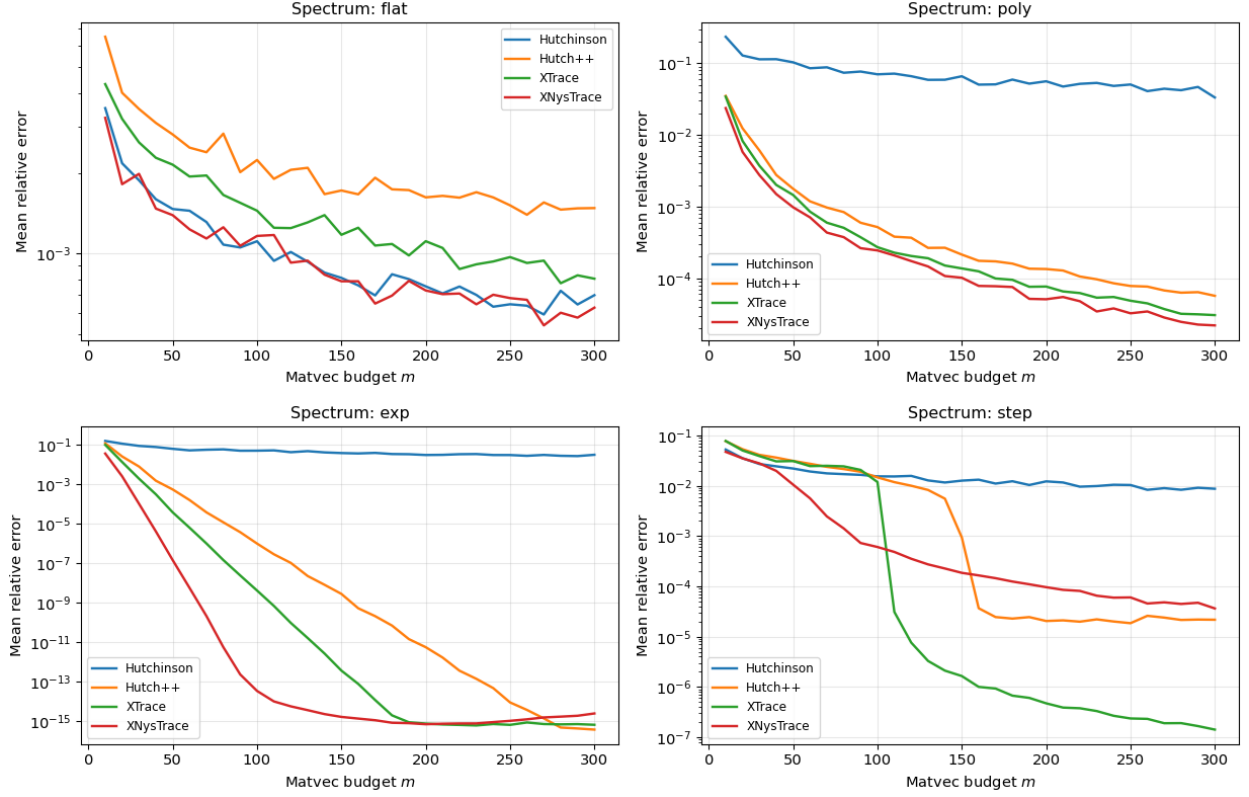


Figure A1: Synthetic Spectra Simulation

with low variance and thus generate effectively flat spectra. By comparison, the BA graph is based around preferential attachment, causing "hub nodes" to be significantly more likely to have additional edges as the network grows, and thus was expected to show spectral decay more akin to the exponential spectra simulated in the original manuscript. The SBM graph was expected to be show a mixture of both spectra with dense communities generating dominant eigenvalues amongst a generally flat spectrum.

While graphs with polynomial and stepwise spectra are also possible, these are less relevant to this discussion for the following reasons. In the polynomial case, all variance reduction techniques work similarly and converge at roughly similar rates, with XNysTrace and XTrace at slightly better performance per matvec. In the case of the stepwise spectra, we can again see that at roughly $m = 50, 100, 150$, the XNysTrace, XTrace, and Hutch++ algorithms break away from the Hutchinson estimator, corresponding with where each method's low rank approximation is approximately equal to 50 where the function steps down. Note also that due to the nuclear norm penalty on XNysTrace as shown in Equation 3, the long eigenvalue tail results in a worse performance than the XTrace algorithm.

Finally, the two trace measures $\text{tr}(M^4)$ and $\text{tr}(e^M)$ were chosen for similar reasons to the ER and BA graphs. The former, which raises eigenvalues to a relatively low polynomial power, is expected to test flat-like behavior and the latter, which raises eigenvalues exponentially, creates extreme concentration and tests exponential type behavior. By comparing any given graph across both regimes, and in particular the mixed-type SBM graph, we expect to demonstrate the advantages of different trace estimation algorithms in different spectral situations.